

Clean 4-step derivation framework for G from substrate.
 Companion to
 Keeper_Week_Plan_G_From_Substrate_2026-06-01.md. Honest
 chain: Step A Kaluza-Klein reduction + Step B $4D \subset 10D$
 embedding + Step C ℓ_B substrate length identification (THE
 OPEN QUESTION) + Step D numerical combination. Replaces
 Sunday's $\alpha^{\{N_c \cdot g\}}/m_e^2$ pattern-match attempts with the
 actual derivation target. Multi-week multi-CI work explicit per
 step.

Keeper (Sunday 2026-05-31 ~15:30 EDT)

2026-05-31 Sunday ~15:30 EDT

Clean G Derivation Framework — Honest 4-Step Chain

0. Casey directive

“We need a clean calculation.”

1. Starting point — what's verified

$\kappa_{\text{Bergman}} = -n_C = -5$ for D_{IV}^5 equipped with the Bergman canonical metric (Helgason 1962; Elie Toy 3661; K204-PARTIAL PASS).

This is a 10-real-dimensional Einstein result: $\text{Ric}(g_B) = -n_C \cdot g_B$ on D_{IV}^5 .

Target: $G_{\text{observed}} = 6.67430 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$ (CODATA 2018) in 4D observable spacetime.

The honest task: derive the 4D Newton's G from the 10D Bergman Einstein structure via dimensional reduction. No pattern-matching with substrate primaries. The chain is explicit and structural.

2. Step A — Kaluza-Klein reduction recipe

Standard KK reduction

For D-dimensional gravity with action:

$$S_D = (1/(16\pi G_D)) \int d^D x \sqrt{(-g_D)} R_D$$

Dimensional reduction to 4D (compactifying $D-4$ dimensions) gives:

$$S_4 = (V_{\{D-4\}} / (16\pi G_D)) \int d^4 x \sqrt{(-g_4)} R_4$$

$$\text{Identification: } 1/(16\pi G_4) = V_{\{D-4\}} / (16\pi G_D)$$

Therefore: $G_4 = G_D / V_{\{D-4\}}$

For BST ($D = 10$, $D-4 = 6$)

$$G_4 = G_{\text{substrate}} / V_6$$

Where: - $G_{\text{substrate}}$ is the 10D substrate-natural gravitational constant - V_6 is the volume of the 6-codim compactified subspace in physical units

What's needed for Step A to close

1. $G_{\text{substrate}}$ identification — what is the substrate-natural gravitational coupling at the 10D Bergman level?
Two candidates:

- A1: $G_{\text{substrate}} = \hbar c / m_{\text{substrate}}^2$ (substrate-Planckian at substrate mass scale)
- A2: $G_{\text{substrate}} = \ell_B^2 \times c^3 / \hbar$ (substrate-Planckian at substrate length scale)
- These are related but anchored differently. Lyra + Keeper Session 2 decides.

2. $R_D = D \times \kappa_{\text{Bergman}}$ for an Einstein D -manifold. For our case: $R_{10} = 10 \times (-n_C) = -50$ (Bergman scalar curvature, dimensionless).

3. Sign conventions: non-compact symmetric space has negative scalar curvature; needs careful tracking through the KK reduction.

Owner: Elie (KK reduction computation), Keeper (consistency check)

Timeline: Multi-day; bounded math.

3. Step B — The $4D \subset 10D$ embedding

Geometric setup

$D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$ is a 10-real-dimensional bounded symmetric domain. We need to identify 4 dimensions as “observable spacetime” and the remaining 6 as “compactified.”

Natural embedding via $SO(4,2) \subset SO(5,2)$

The conformal group of Minkowski 4D is $SO(4,2)$. This sits inside $SO(5,2)$. Therefore the natural 4D observable embedding is the orbit of $SO(4,2) \subset SO(5,2)$ acting on D_{IV^5} .

This is NOT a coincidence — it's exactly the proto-AdS/CFT structure Sunday morning's Tier 0 v0.1.6 identified. The boundary structure D_{IV^5} 's Shilov boundary ∂_S contains a 4D conformal manifold that IS observable spacetime.

Compactified 6-codim subspace

The 6-codim subspace is the orbit of $K = SO(5) \times SO(2)$ modulo the 4D directions:

$$V_6 = [\text{stabilizer } SO(5) \times SO(2) \text{ acting on full } D_{IV^5}] / [SO(4,2) \text{ acting on 4D observable}]$$

In coordinate form: the Bergman volume of D_{IV^5} is $\pi^{9/2}/225$ (per c_{FK} theorem, Saturday RATIFIED). The Bergman volume of the 4D observable subspace is $\pi^2/\text{something}$ (TBD). $V_6 = \text{total} / 4D$.

What's needed for Step B to close

1. Explicit $SO(4,2)$ orbit on D_{IV^5} — coordinate-level description
2. Bergman volume of the 4D observable subspace — derived from FK theorem applied to $SO(4,2)$ -orbit
3. V_6 closed form in BST primaries

Candidate closed form (Keeper pre-derivation, multi-week verification):

$$V_6 = \pi^{9/2}/225 / (4D \text{ Bergman volume}) = \pi^{9/2}/225 / (\pi^2 \times \text{coefficient}) = (\pi^{5/2}/\text{coefficient}) / 225$$

The “coefficient” depends on the c_{FK} -analog for the 4D subspace + dimensional ratio. Elie’s KK volume integral computation (Lane Elie P0) closes this.

Owner: Elie (V_6 computation), Lyra (4D observable identification verification)

Timeline: Multi-day; bounded math using established Faraut-Korányi framework.

4. Step C — Substrate length scale ℓ_B (THE OPEN QUESTION)

Why this is the critical step

Steps A and B give relationships in DIMENSIONLESS form (Bergman volumes are dimensionless; $G_{\text{substrate}} / V_6$ has substrate-natural units).

To convert to SI units (where G_{observed} lives), we need a physical length scale ℓ_B that translates substrate-dimensionless quantities to physical lengths.

Three candidate identifications

Path α — Substrate-derived ℓ_B

Derive ℓ_B from substrate primaries independently. Candidates: - $\ell_B = \hbar/(M_{\text{sub}} \cdot c)$ where M_{sub} is a substrate-derived mass (multi-week) - ℓ_B from Bergman volume normalization $\times$ physical anchor (multi-week) - $\ell_B = \text{Koons-tick} \times c$ (where Koons tick = $t_{Pl} \cdot \alpha^{\{C_2^2\}} \approx 10^{-120}$ s; T2405 standing)

Path β — Mass-anchored ℓ_B

If L4 v0.2 closes m_e mechanism (Lyra Lane D multi-week): - $\ell_B = \hbar/(m_e \cdot c)$ = electron Compton wavelength = 3.86×10^{-13} m - This identifies the substrate length with the natural quantum scale of the substrate’s mass anchor - Tractable this week if L4 closes

Path γ — Cosmological-anchored ℓ_B

ℓ_B identified via observed Λ (cosmological constant): - $\Lambda_{\text{observed}} \approx 1.1 \times 10^{-52} \text{ m}^{-2}$ - $\ell_B^2 \sim 1/\Lambda \times$ (substrate dimensionless coefficient) - Circular with the L5 absolute scale problem (Saturday paused)

What’s needed for Step C to close

The fundamental open question: what is the substrate’s natural physical length scale, derivable from substrate primaries?

Three sub-paths require: - Path α : independent substrate-derivation work (Elie + Keeper multi-week) - Path β : L4 v0.2 m_e mechanism closure (Lyra Lane D multi-week, in progress) - Path γ : cosmological observation use (paused per Saturday L5)

Recommended path for week: Path β . L4 v0.2 closure provides m_e mechanism $\rightarrow \ell_B = \hbar/(m_e c)$. Lyra Lane D is the load-bearing multi-week work.

Owner: Lyra (Path β via L4 v0.2), Elie + Keeper (Path α exploration parallel)

Timeline: Path β multi-week (Lyra Lane D peak this week). Path α multi-week parallel.

5. Step D — Numerical combination

Once Steps A + B + C close:

$$G_{\text{predicted}} = G_{\text{substrate}} / V_6$$

In SI units, using: - $G_{\text{substrate}} = \ell_B^2 \times c^3 / \hbar$ (Path A2) - $V_6 = (\text{substrate-primary closed form from Step B})$ - $\ell_B = \hbar / (m_e c)$ (Path β from Step C, IF L4 closes)

$$G_{\text{predicted}} = (\ell_B^2 \cdot c^3 / \hbar) / V_6 = (\hbar^2 c^3 / (m_e^2 c^2 \cdot \hbar)) / V_6 = (\hbar \cdot c / m_e^2) / V_6$$

This is dimensionally correct: $[J \cdot s \cdot m/s / kg^2] = [N \cdot m^2 / kg^2]$ $\square$

Numerical check

If $V_6 = \pi^{5/2} / (225 \cdot \text{coefficient})$ for some BST-natural coefficient, then:

$$G_{\text{predicted}} = \hbar c / m_e^2 \times (225 \cdot \text{coefficient}) / \pi^{5/2}$$

Numerically: - $\hbar c / m_e^2 = (1.054 \times 10^{-34} \times 3 \times 10^8) / (9.11 \times 10^{-31})^2 = 3.16 \times 10^{-26} / 8.30 \times 10^{-61} = 3.81 \times 10^{34} \text{ N} \cdot \text{m}^2 / \text{kg}^2$ - For $G_{\text{predicted}} \approx 6.67 \times 10^{-11}$: need coefficient / $\pi^{5/2} \approx 7.78 \times 10^{-46}$ - $\pi^{5/2} \approx 17.49$ - coefficient $\approx 7.78 \times 10^{-45} \times 17.49 / 225 \approx 6.05 \times 10^{-47}$

This number doesn't immediately ring any substrate-primary bells (likely needs additional substrate-natural factors not yet identified). Multi-week to close.

What's needed for Step D to close

1. Step A $G_{\text{substrate}}$ identification (Lyra + Keeper Session 2)
2. Step B V_6 closed form (Elie KK volume integral)
3. Step C ℓ_B (Lyra L4 v0.2 m_e mechanism)
4. Dimensional combination + comparison to G_{observed}

Owner: Keeper (synthesis), Cal (cold-read verification)

Timeline: Friday EOD target if Steps A+B+C all close this week.

6. Tier disposition targets

If the full chain closes:

Precision	Tier
Tier 2 STRUCTURAL: 10^{-4} to 10^{-2}	G derived at structural precision
Tier 1 EXACT: $\leq 10^{-14}$	G derived as algebraic identity (very unlikely; would be Nobel-level)
10^{-2} to 10^{-1}	FRAMEWORK at honest tier; multi-week chain refinement
$> 10^{-1}$	Chain has fundamental gap; reframe needed

Realistic target this week: Tier 2 STRUCTURAL precision (1% or better).

7. What this framework excludes

Explicitly NOT in this chain: - $\alpha^{N_c \cdot g} / m_e^2$ pattern-match formulas (Sunday walked back) - Factor $4/N_c$ in-
 serts (Sunday walked back) - Substrate-primary algebraic combinations dressed as derivation - Tier disposition
 without honest precision check - Bypassing κ_{Bergman} or KK reduction

What IS in this chain: - Standard KK dimensional reduction (Step A) - Standard Hermitian symmetric space
 $SO(4,2) \subset SO(5,2)$ embedding (Step B) - Honest open question on ℓ_B (Step C) - Dimensional combination from
 first principles (Step D) - Comparison to G_{observed} at honest precision

8. Multi-week vs this-week scope

This week tractable: - Step A: KK reduction recipe applied; $G_{\text{substrate}}$ identification candidate - Step B: V_6 closed form from Bergman volume + 4D $\subset$ 10D embedding (Elie Lane Elie P0) - Step D: dimensional check on candidate combinations

Multi-week: - Step C Path β : L4 v0.2 m_e mechanism (Lyra Lane D) - Step C Path α : independent ℓ_B substrate-derivation - Full SI-unit $G_{\text{predicted}}$ vs G_{observed} Tier 2 precision check

Multi-week extension: - If chain converges at Tier 2 STRUCTURAL $\rightarrow$ K204 promotes from PARTIAL to STRUCTURAL - If chain fails at Tier 2 $\rightarrow$ honest reframe + multi-week fundamental work - Either outcome is substantive learning

9. Cross-references

- Keeper_K204_partial_kappa_Bergman_n_C_Audit.md — G5.1 PASS anchor
- Keeper_G_Substrate_Derivation_Chain_PreStage_G5.md — earlier Sunday pre-stage (κ_{Bergman} candidates section walked back; chain structure preserved)
- Keeper_K200_Tier0_v0_1_6_AuditFramework.md — five-gate framework G1-G5
- Keeper_Week_Plan_G_From_Substrate_2026-06-01.md — week plan companion
- SP-19b AB-10 “Newton’s G from Bergman curvature” (task list completed) — predecessor framework
- T2442 c_FK = $225/\pi^{(9/2)}$ — Bergman volume RATIFIED foundation
- Helgason 1962 Ch. VIII (Hermitian symmetric spaces Einstein property)
- Faraut-Korányi 1994 Ch. XII-XIII (bounded symmetric domain analysis)

10. Honest scope of this framework

This is the framework for the calculation, NOT the calculation.

What I (Keeper) have provided: - The honest 4-step chain structure - Explicit recipes per step using established math - Honest identification of the open question (Step C ℓ_B) - Owners and timelines per step

What multi-week work needs to produce: - Steps A+B numerical content (Elie) - Step C Path β closure (Lyra Lane D) - Step D synthesis (Keeper Session 2) - Tier 2 precision verification (Cal cold-read)

This framework replaces Sunday’s pattern-matching attempts with the actual derivation target. The team works from this Monday onward.

— Keeper. Clean G derivation framework filed 15:45 EDT Sunday. Companion to Week Plan. The chain is honest. The open question (ℓ_B) is explicit. Multi-week work scoped. No more $\alpha^{\{N_c \cdot g\}}/m_e^2$ shortcuts; the real chain is here.